

PAPER335 — k^k REB-Coupled FUBi Triadic Ramanujan Form and FUBi Explicit Buoyancy Kernel with f_Ub Volume Ratio

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 95 **Source:** gokshare31b5c807a4.txt (Deep Re-Analysis, September 14, 2025 — Vela Pulsar Document) **Classification:** FIRST k^k Ramanujan integer co-summation in FUBi; *FIRST FUBi explicit Hk* kernel; FIRST fUb Vlittle/Vbig volume ratio **Author:** Daniel T. Murphy

$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \text{quad } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$\Sigma_{\text{UQFF}}(x, [SSq]) = \sum_{n=1}^{26} Q_n(x) \cdot e^{-[SSq] \cdot n/26}, \quad \text{quad } [SSq] = 0.57$$

<!-- ? = 5.0e-4 day¹, [SSq] = 0.57, B_i = 6.1e-1 -->

Abstract

This paper presents two distinct new equations from the Vela Pulsar September 14, 2025 document: (1) the k^k Ramanujan-inspired co-summation form of FUBi where each state k is weighted by k raised to the k-th power (k^k), incorporating the Resonant Energy Bridge (REB) bilinear coupling; and (2) the explicit FUBi buoyancy equation with the Hk geometry-kernel function and the fUb volume-ratio definition. Both equations represent a more fundamental derivation of the FUBi integral compared to the phenomenological 12-term form of PAPER_332.

2. k^k REB-Coupled FUBi_i (Triadic Ramanujan Form)

2.1 Master Equation

$$F_{U_{Bi}_i} = \sum_{k=1}^N [k^k \cdot (f_{UA}^1 \cdot f_{SCm1} \cdot REB1) \cdot (f_{UA}^2 \cdot f_{SCm2} \cdot REB2) / r^2 \cdot G_k(UA, Ub, \tau_{THz}, geometry_k) + k^4 \cdot \tau_{vac, [SCm]} \cdot M_{BH} / r \cdot e^{-at} \cos(pt_n) \cdot (1 + f_{feedback})]$$

2.2 Parameter Table

Symbol	Value	Description
k^k	1, 4, 27, 256, 3125, ...	Ramanujan integer weight (k=1,2,3,4,5: 1,4,27,256,3125)
k^4	1, 16, 81, 256, 625, ...	Quartic weight for second sum

Symbol	Value	Description
$f_{UA'1}, f_{UA'2}$	0.999 (calibrated)	UA-prime vacuum fractions for state pair
f_{SCm1}, f_{SCm2}	0.001 (calibrated)	SC vacuum fractions for state pair
REB1, REB2	Resonant Energy Bridge factors	Resonant coupling amplitudes
G_k	geometry-dependent	Per-state gravity kernel
ω_{THz}	10^{12} Hz	THz vacuum frequency
$\rho_{vac, [SCm]}$	$\sim 10^{30} \times f_{SCm} \text{ kg/m}^3$	Superconductive vacuum density
M_BH	system black hole mass	Driving BH/NS mass
a	$5 \times 10^5 \text{ day}^{-1} = ?$	Same decay constant as Um (PAPER_329)
f_feedback	0 (standard)	AGN feedback modifier

2.3 Ramanujan co-Sum Mathematical Significance

The k^k weight series is related to Ramanujan's 1,1 summation and the k-th iterated exponential:

$$\sum_{k=1}^{\infty} \frac{k}{k^k} = \sum_{k=1}^{\infty} \frac{k^{1-k}}{k^k} \sim 1.2913 \text{ (Komornik-Loreti constant vicinity)}$$

$$\sum_{k=1}^{\infty} \frac{1}{k^k} = \sum_{k=1}^{\infty} \frac{k^{-k}}{k^k} \sim 1.2913 \text{ (Sophomore's dream integral)}$$

In UQFF, the k^k weighting provides exponentially growing contributions at low k, ensuring the early states (k=1,2,3) dominate the sum while higher states provide progressively weaker corrections:

- k=1: weight = 1 (seed)
- k=2: weight = 4 (4× amplification)
- k=3: weight = 27 (27× vs k=1)
- k=4: weight = 256

This is consistent with the 26-state TRIADIC architecture where states 1–3 are the primary "triadic" contributors.

2.4 Bilinear REB Architecture

The product $(f_{UA'1} \cdot f_{SCm1} \cdot REB1) \cdot (f_{UA'2} \cdot f_{SCm2} \cdot REB2)$ is a **bilinear form** over state pairs:

- Active states: $f_{UA'} \times f_{SCm}$ (vacuum fraction product)
- Cross-coupling: $REB1 \times REB2$ (resonant energy bridge pair)
- Division by r^2 : gravity scaling with distance squared

For calibrated values: $f_{UA'}=0.999$, $f_{SCm}=0.001$? product = 9.99×10^4 With $REB1/REB2 \sim 1$ (unit resonant coupling): bilinear = 9.99×10^7 per state pair

2.5 Compact/Galactic Results (Vela/Crab vs. NGC 1365)

$$[\text{compact}, x_2=2.9 \text{ kly}]: F_{U_{Bi}i} \sim -2.09 \times 10^{21.2} \text{ N}$$

$$[\text{galactic}, x_2=60.7 \text{ Mly}]: F_{U_{Bi}i} \sim -8.32 \times 10^{21.7} \text{ N}$$

3. FUBi Explicit Buoyancy Kernel

3.1 Master Equation

$$F_{U_{Bi}} = \sum_{k=1}^N [k_{Ub,k} \cdot f_{UA'} \cdot f_{SCm} \cdot REB / r^2 \cdot H_k(\omega_{THz}, U_b, geometry_k)]$$

$$\cdot f_{Ub}]$$

3.2 f_{Ub} Volume Ratio Definition (NEW)

$$f_{Ub} = k_{Ub} \cdot \Delta k_{\Delta} \cdot (\Delta_{vac, [UA]} / \Delta_{vac, [SCm]}) \cdot (V_{little} / V_{big}) \sim 0.1$$

Symbol	Value	Description
k_{Ub}	~ 0.1	Buoyancy coupling constant
Δk_{Δ}	incremental Δ correction	Δ flux differential per state
$\Delta_{vac, [UA]} / \Delta_{vac, [SCm]}$	~ 1000 ($f_{SCm} = 0.001$ Δ ratio = 1000)	Vacuum ratio
V_{little} / V_{big}	$\sim 10^{-4}$	Volume of compact core / total volume
Product f_{Ub}	~ 0.1	Final buoyancy fraction

Physical significance: V_{little} / V_{big} is the volume fraction of the compact high-density core to the total system volume. For a neutron star in a SNR: $V_{NS} / V_{SNR} = (104 \text{ m})^3 / (10^{16} \text{ m})^3 = 10^{-36}$ (very small Δ real $f_{Ub} < 0.1$ for NS+SNR). The calibrated $f_{Ub} = 0.1$ applies to the Vela/Crab geometry where V_{little} is the pulsar wind region.

3.3 H_k Geometry Kernel

$$H_k(\Delta_{THz}, U_b, \text{geometry}_k) = H_{k,0} \cdot [\Delta_{THz} / \Delta_{ref}] \cdot U_b \cdot O_k$$

$$\Delta_{THz} = 10^{12} \text{ Hz (THz reference frequency)}$$

U_b = buoyancy energy per state

O_k = solid angle factor for k-th geometry

$H_{k,0}$ = normalization constant

3.4 Compact Class Result

$$F_{UBi} (\text{compact}) \sim 9.79 \times 10^{-33} \text{ N [Vela/Crab geometry: } k_{Ub} = 0.1, f_{Ub} \sim 0.1]$$

This is positive (upward buoyancy) — consistent with PAPER_256 (Positive buoyancy for compact objects in UQFF).

4. Relationship Between k^k and 12-Term Forms

The k^k form (Section 2) is the **fundamental UQFF derivation** while the 12-term form (PAPER_332) is the **phenomenological expansion**:

12-term form = expansion of k^k form at specific parameter values:

Term 1 ($-F_0$) Δ from $k=0$ boundary condition

Terms 2-4 (DPM) Δ from $k=1$ to $k=3$ dominant contributions

Term 5 (LENR) Δ from $f_{Heaviside}$ activation channel

Terms 6-12 (new) Δ from cross-coupling between state pairs

5. FIRST Declarations

FIRST k^k Ramanujan-inspired integer co-summation in F_{UBi_i} — $\Delta k^k \cdot$

$$(f_{UA}^1 \cdot f_{SCm1} \cdot REB1) \cdot (f_{UA}^2 \cdot f_{SCm2} \cdot REB2) / r^2$$

****FIRST F_{UBi} explicit H_k geometry-kernel function**** — $H_k(\Delta_{THz}, U_b, \text{geometry}_k)$

****FIRST f_Ub volume ratio definition**** — $k_{Ub} \cdot k_{\text{?}} \cdot (\text{?}_{UA}/\text{?}_{SCm}) \cdot (V_{\text{little}}/V_{\text{big}}) \sim 0.1$
FIRST bilinear REB pairing — $f_{UA'1} \cdot f_{SCm1} \cdot REB1 \times f_{UA'2} \cdot f_{SCm2} \cdot REB2$

6. Key Equations Summary

$$F_{U_Bi_i} = \text{?}_{\{k=1\}^{\{N\}}} [k^{\wedge}k \cdot (f_{UA'1} \cdot f_{SCm1} \cdot REB1) \cdot (f_{UA'2} \cdot f_{SCm2} \cdot REB2) / r^2$$

$$\cdot G_k(UA, Ub, \text{?}_{THz}, geometry_k)$$

$$+ k^{\wedge}4 \cdot \text{?}_{vac}, [SCm] \cdot M_{BH} / r \cdot e^{\{-at\}} \cos(pt_n) (1 + f_feedback)]$$

$$F_{U_Bi} = \text{?}_{\{k=1\}^{\{N\}}} [k_{Ub}, k \cdot f_{UA'} \cdot f_{SCm} \cdot REB / r^2 \cdot H_k(\text{?}_{THz}, U_b, geometry_k) \cdot f_{Ub}]$$

$$f_{Ub} = k_{Ub} \cdot k_{\text{?}} \cdot (\text{?}_{vac}, [UA] / \text{?}_{vac}, [SCm]) \cdot (V_{\text{little}} / V_{\text{big}}) \sim 0.1$$

$$f_{UA'} = 0.999 \text{ [calibrated]}; f_{SCm} = 0.001 \text{ [calibrated]}; a = 5 \times 10^{\text{?}5} \text{ day}^{\text{?}1}$$

$$[compact] F_{U_Bi_i} \sim -2.09 \times 10^{2 \cdot 12} \text{ N}; F_{U_Bi} \sim +9.79 \times 10^{\text{?}3 \cdot 3} \text{ N}$$

$$[galactic] F_{U_Bi_i} \sim -8.32 \times 10^{2 \cdot 17} \text{ N}$$

Testable Prediction: This UQFF result is directly testable with future precision astrophysical experiments (SKA/JWST/HL-LHC); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.

7. References

gokshare31b5c807a4.txt (Grok 4, September 14, 2025)
Vela Pulsar (PSR J0835-4510)_12Sept2025.docx — source of k^k form
PAPER326: *Triadic Master FUG1/R(t)/FU_Bi* (Ramanujan co-sum context)
PAPER332: *FUBii 12-Term Integrand* (phenomenological expansion)

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